Supplementary Material for "FundusShift: Benchmarking Accuracy and Reliability of Fundus Disease Classification at Unseen Sources, and a Low-Capacity Remedy"

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This supplement gives every table behind the main paper, plus analyses that did not fit there. All tables are generated from the saved results in the repository (`https://doi.org/10.5281/zenodo.23911966`) by `paper/make_assets.py` and `paper/make_supp.py`; numbers are means over seeds unless stated. Abbreviations: bAcc, balanced accuracy; Eq-src, mean of per-source bAcc; R/T/F, raw, temperature scaling (TS), focal temperature scaling (FTS); LP, linear probe; LAC, least-ambiguous conformal sets; G→N, glaucoma images accepted and predicted normal, over all glaucoma images.

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S1 Data

S1.1 Sources, label scope and licences

Table S1 lists every source with its class counts per split. "Scope" records which diseases a source graded: in DR-only sources a normal label means "no DR", and glaucoma was not assessed.

Table S1: Sources. Counts are N/D/G. Development sources contribute train, validation and calibration; test sources and Chákṣu contribute only the evaluation split.

Source	Country	Scope	Licence	Train	Validation	Calibration	Test or OOD
Bangladesh Eye	Bangladesh	full	CC BY 4.0	677/1200/925	85/150/116	84/150/115	0/0/0
OIA-DDR	China (147 hospitals, 23 pro	DR only	CC BY-NC-SA 4.0	5012/5004/0	627/626/0	626/625/0	0/0/0
DeepDRiD	China	DR only	CC BY-SA 4.0	732/868/0	94/110/0	88/108/0	0/0/0
GRAPE	China	full	CC0	0/0/470	0/0/58	0/0/58	0/0/0
IDRiD	India	DR only	CC BY 4.0	133/277/0	17/35/0	17/35/0	0/0/0
Jichi	Japan	DR only	CC BY 4.0	5028/2504/0	635/308/0	632/308/0	0/0/0
SUSTech-SYSU	China	DR only	CC BY 4.0	505/470/0	63/59/0	63/59/0	0/0/0
AFIO	Pakistan (Armed Forces Insti	full	CC BY 4.0	–	–	–	67/18/44
FIVES	China	full	CC BY 4.0	–	–	–	193/200/193
HRF	Germany / Czech Republic	full	CC BY 4.0	–	–	–	15/15/15
HYGD	Israel	full	ODC-By 1.0	–	–	–	0/0/548
PALM	China (Zhongshan Ophthalmic	full	CC0	–	–	–	161/0/0
PAPILA	Spain	full	GPL-3.0	–	–	–	333/0/87
Paraguay	Paraguay	DR only	CC BY 4.0	–	–	–	708/723/0
Chákṣu	India	full	CC BY 4.0	–	–	–	1156/0/187

Reading. The development side holds 1,395 glaucoma training images, all from Bangladesh Eye (925) and GRAPE (470), whereas DR and normal images come from all seven sources. Six of seven development sources graded only DR. On the test side, only Paraguay is DR-only, HYGD is G-only and PALM is N-only, so per-source bAcc on those two reduces to one recall. These facts explain much of the glaucoma behaviour in the main paper.

S1.2 Mapping from source labels

Table S2: Native labels mapped to each harmonized class (first token of the stored label; device and vote suffixes omitted).

Source	Class	Native labels
Bangladesh Eye	N	Healthy
Bangladesh Eye	D	Diabetic_Retinopathy
Bangladesh Eye	G	Glaucoma
OIA-DDR	N	ddr_0
OIA-DDR	D	ddr_1, ddr_2, ddr_3, ddr_4
DeepDRiD	N	icdr_0
DeepDRiD	D	icdr_1, icdr_2, icdr_3, icdr_4
GRAPE	G	glaucoma_follow_up
IDRiD	N	icdr_0
IDRiD	D	icdr_1, icdr_2, icdr_3, icdr_4
Jichi	N	davis_ndr
Jichi	D	davis_pdr, davis_ppdr, davis_sdr
SUSTech-SYSU	N	icdr_0
SUSTech-SYSU	D	icdr_1, icdr_2, icdr_3, icdr_4, laser_scars(reclassified_icdr_2), laser_scars(reclassified_icdr_3), ...
AFIO	N	macula_healthy, od_healthy
AFIO	D	macula_ci-dme
AFIO	G	od_glaucoma
FIVES	N	normal
FIVES	D	diabetic_retinopathy
FIVES	G	glaucoma
HRF	N	healthy
HRF	D	diabetic_retinopathy
HRF	G	glaucoma
HYGD	G	GON+
PALM	N	normal
PAPILA	N	healthy
PAPILA	G	glaucoma
Paraguay	N	1._No_DR_signs
Paraguay	D	2._Mild_(or_early)_NPDR, 3._Moderate_NPDR, 4._Severe_NPDR, 5._Very_Severe_NPDR, 6._PDR, 7._Advanced_PDR
Chákṣu	N	normal
Chákṣu	G	glaucoma_suspect

Reading. A native label is mapped only when it supports the class. Grading scales differ: ICDR grades 1–4, Davis SDR–PDR and named DR categories all become D. Glaucoma suspects in Chákṣu follow the source's majority vote.

S1.3 Leakage control and audit

Duplicates were detected by exact SHA-256 hashes and by a vessel-pattern signature (band-pass green channel, 32×32) with shift-tolerant normalized cross-correlation at threshold 0.70, then joined with patient identifiers into leakage groups. An automated audit of the released splits found zero violations for each of: a leakage group in more than one split; an identical file in more than one split; a patient in more than one split; a near-duplicate pair across splits; an acquisition family on both development and test sides; a source on both sides; a calibration image without a Cal-A/Cal-B assignment. Brightness-altered copies of GRAPE photographs inside the FIVES glaucoma class were found this way and removed. Of 40,414 inventoried images, 35,280 enter the protocol.

Excluded collections. RFMiD 2.0 was used for inventory and duplicate checks only, because class is confounded with camera. A glaucoma collection of optic-disc crops was excluded for its different field of view. OIA-DDR is CC BY-NC-SA 4.0, so derived data remain non-commercial.

Ethics. All images are publicly released; this is a secondary analysis. Ethics and consent statements were found in the publications of FIVES, GRAPE, PAPILA, SUSTech-SYSU, DeepDRiD, Jichi, HYGD, Chákṣu, PALM and Paraguay. Statements were not found on the release pages of HRF, Bangladesh Eye and AFIO. Age and sex are available only for PAPILA, so subgroup fairness cannot be assessed.

S2 Models, Recipes and Prompts

Table S3. Encoders. "Fine-tuned" marks the 13 architectures trained end to end; every encoder was also probed frozen.

Encoder	Pretraining data	Fine-tuned
deit_b16	ImageNet-1k	yes
convnext_t	ImageNet-1k	yes
swin_t	ImageNet-1k	yes
inception_v3	ImageNet-1k	yes
efficientnet_b0	ImageNet-1k	yes
densenet121	ImageNet-1k	yes
vit_b16_in21k	ImageNet-21k	yes
resnet50	ImageNet-1k	yes
mobilenetv3_l	ImageNet-1k	yes
vgg16	ImageNet-1k	yes
vit_b16_mae	ImageNet-1k (MAE, self-supervised)	yes
dinov2_b14	LVD-142M (self-supervised)	no
retfound_mae	MEH-MIDAS + public fundus (MAE, RETFound)	yes
retfound_dinov2	MEH-MIDAS fundus (DINOv2, RETFound)	yes
clip_b16	OpenAI web image-text	no
metaclip_b16	MetaCLIP (web image-text)	no
siglip_b16	WebLI	no
siglip_l16	WebLI	no
siglip2_b32	WebLI	no
siglip2_b16	WebLI	no
siglip2_l16	WebLI	no
bioclip_b16	iNat/biodiversity (TreeOfLife-10M)	no
bioclip2_l14	biodiversity (TreeOfLife-200M)	no
biomedclip	PMC-15M biomedical figures	no
medsiglip	Medical image-text (MedGemma report); fundus: EyePACS	no
flair	Retinal image-text; includes five FundusShift test sources	no

Table S5: Prompt banks used by every zero-shot model and by LANTERN's text branch.

Bank	Prompts
CLIP ensemble	a photo of .; a colour fundus photograph of .; a retinal image of .; a fundus photo showing .; an ophthalmology image of .; a close-up retinal photograph of .; . with class names: N: a normal retina, D: diabetic retinopathy, G: glaucoma
CuPL (N)	a healthy fundus with a pink optic disc, a clear macula and regular retinal vessels. a normal retina without haemorrhages, exudates or optic disc abnormality. a fundus photograph with an evenly coloured background and a well-defined foveal reflex. a retina with a small physiological optic cup and an intact neural rim. a fundus free of microaneurysms, cotton-wool spots and neovascularisation.
CuPL (D)	a retina with scattered microaneurysms and dot-blot haemorrhages. a fundus with yellow hard exudates near the macula. a retinal photograph with cotton-wool spots and venous beading. a fundus with abnormal new vessels and preretinal haemorrhage. a diabetic eye with intraretinal haemorrhages in several quadrants.

Table S5: Prompt banks used by every zero-shot model and by LANTERN's text branch.

Bank	Prompts
CuPL (G)	an optic disc with a large cup-to-disc ratio. a fundus with a thinned or notched neural rim of the optic disc. a retina with retinal nerve fibre layer defects next to the optic disc. an optic nerve head with vertical cupping and pallor. a fundus with peripapillary atrophy and glaucomatous optic neuropathy.
Single (N)	a colour fundus photograph of a healthy retina with a normal optic disc and no lesions.
Single (D)	a colour fundus photograph with diabetic retinopathy showing microaneurysms, haemorrhages and exudates.
Single (G)	a colour fundus photograph with glaucoma showing an enlarged optic cup and a thinned neural rim.

S3 Complete Benchmark Results

Table S6: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

Setting	Encoder	Config	Val	Test	Eq-src	Chákṣu	ECE R/T/F	NLL R/T/F
16-shot probe	flair†	none/mbls	.809	.750	.684	.683	.134/.128/.127	.947/.915/.891
16-shot probe	medsiglip	none/ce	.802	.748	.683	.675	.096/.044/.037	.701/.644/.644
16-shot probe	siglip2_l16	none/mbls	.696	.621	.564	.195	.124/.075/.074	.999/.885/.889
16-shot probe	biomedclip	none/ce	.677	.479	.430	.450	.198/.113/.120	1.18/1.05/1.06
16-shot probe	dinov2_b14	none/ce	.675	.529	.503	.192	.275/.080/.088	1.53/.993/.996
16-shot probe	efficientnet_b0	none/ce	.665	.482	.458	.279	.216/.081/.085	1.24/1.02/1.03
16-shot probe	retfound_mae	none/mbls	.662	.373	.339	.207	.437/.238/.241	1.94/1.17/1.17
16-shot probe	siglip_l16	none/ce	.658	.440	.414	.316	.314/.166/.168	1.51/1.16/1.16
16-shot probe	siglip_b16	none/ce	.653	.526	.516	.266	.175/.093/.102	1.24/1.02/1.03
16-shot probe	densenet121	none/ce	.653	.514	.499	.347	.207/.063/.056	1.15/.963/.955
16-shot probe	convnext_t	none/ce	.649	.523	.520	.403	.214/.063/.069	1.24/.971/.970
16-shot probe	clip_b16	none/mbls	.649	.486	.419	.251	.192/.103/.108	1.24/1.08/1.09
16-shot probe	deit_b16	none/mbls	.645	.517	.484	.233	.228/.096/.097	1.24/1.01/1.01
16-shot probe	retfound_dinov2	none/ce	.644	.437	.426	.158	.282/.087/.095	1.47/1.09/1.09
16-shot probe	mobilenetv3_l	none/ce	.643	.450	.426	.357	.277/.106/.113	1.48/1.08/1.09
16-shot probe	siglip2_b16	none/ce	.638	.456	.422	.281	.315/.137/.141	1.85/1.11/1.11
16-shot probe	swin_t	none/ce	.634	.494	.458	.204	.214/.059/.062	1.21/1.00/1.01
16-shot probe	vit_b16_in21k	none/ce	.630	.495	.469	.268	.234/.087/.090	1.34/1.02/1.02
16-shot probe	resnet50	none/mbls	.628	.541	.509	.429	.149/.070/.061	1.12/.986/.979
16-shot probe	inception_v3	none/mbls	.627	.457	.425	.437	.323/.099/.101	1.59/1.07/1.08
16-shot probe	siglip2_b32	none/ce	.622	.471	.439	.375	.260/.098/.098	1.41/1.07/1.07
16-shot probe	vit_b16_mae	none/ls	.621	.392	.412	.239	.331/.189/.200	1.57/1.20/1.20
16-shot probe	vgg16	none/ls	.619	.468	.459	.378	.213/.135/.125	1.73/1.24/1.19
16-shot probe	metaclip_b16	none/ce	.613	.612	.539	.458	.145/.085/.079	1.10/.906/.903
16-shot probe	bioclip2_l14	none/ce	.612	.495	.467	.407	.247/.064/.068	1.48/1.02/1.03
16-shot probe	bioclip_b16	none/ce	.597	.411	.386	.204	.347/.142/.141	1.64/1.14/1.14
CoOp	siglip_b16	ce	.660	.491	.458	.318	.236/.175/.177	1.35/1.18/1.16
CoOp	biomedclip	ls	.654	.475	.437	.379	.235/.163/.173	1.37/1.19/1.21
CoOp	clip_b16	ls	.631	.448	.382	.206	.235/.189/.190	1.27/1.16/1.17
CoOp	bioclip2_l14	mbls	.622	.474	.462	.270	.235/.133/.134	1.50/1.21/1.24
CoOp	siglip_l16	ce	.612	.403	.479	.368	.107/.189/.191	1.13/1.25/1.28
CoOp	bioclip_b16	mdca	.593	.415	.422	.152	.429/.154/.155	2.59/1.19/1.20
CoOp	metaclip_b16	ls	.585	.540	.483	.358	.125/.089/.084	.990/.951/.944
CoOp	siglip2_l16	zs_penalty	.408	.459	.489	.444	.099/.044/.045	1.01/.994/.997
CoOp	siglip2_b32	zs_penalty	.401	.399	.315	.427	.045/.191/.186	1.05/1.11/1.10
CoOp	siglip2_b16	zs_penalty	.358	.335	.233	.205	.132/.224/.224	1.10/1.16/1.16
CoOp ctx-init	biomedclip	mdca	.642	.497	.447	.384	.287/.150/.160	1.76/1.19/1.22
CoOp ctx-init	clip_b16	mdca	.638	.499	.429	.171	.189/.133/.134	1.20/1.08/1.08
CoOp ctx-init	metaclip_b16	mdca	.634	.605	.534	.418	.092/.043/.042	.884/.838/.836

Table S6: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

Setting	Encoder	Config	Val	Test	Eq-src	Chákṣu	ECE R/T/F	NLL R/T/F
CoOp ctx-init	bioclip2_l14	mdca	.611	.468	.458	.231	.188/.115/.115	1.36/1.18/1.20
CoOp ctx-init	bioclip_b16	mbls	.583	.444	.418	.157	.419/.173/.175	2.56/1.21/1.22
CoOp ctx-init	siglip_l16	zs_penalty	.442	.421	.519	.480	.075/.092/.097	1.08/1.09/1.09
CoOp ctx-init	siglip_b16	mdca	.421	.458	.513	.492	.048/.120/.119	1.09/1.06/1.06
CoOp ctx-init	siglip2_l16	mdca	.386	.361	.450	.499	.231/.183/.178	1.29/1.20/1.23
CoOp ctx-init	siglip2_b32	zs_penalty	.358	.380	.447	.518	.054/.061/.058	1.06/1.06/1.06
CoOp ctx-init	siglip2_b16	focal	.351	.341	.424	.496	.178/.157/.159	1.24/1.20/1.22
CoOp + LP	siglip_b16	mbls	.668	.496	.463	.314	.243/.176/.179	1.37/1.18/1.17
CoOp + LP	biomedclip	mdca	.652	.482	.433	.368	.292/.177/.186	1.68/1.22/1.24
CoOp + LP	siglip_l16	ce	.647	.447	.490	.327	.101/.169/.172	1.12/1.23/1.27
CoOp + LP	clip_b16	ls	.637	.449	.384	.213	.244/.190/.191	1.29/1.16/1.16
CoOp + LP	siglip2_l16	ls	.632	.559	.543	.276	.154/.030/.030	.969/.897/.895
CoOp + LP	metaclip_b16	mbls	.628	.594	.506	.410	.170/.067/.066	.959/.860/.857
CoOp + LP	bioclip2_l14	mbls	.623	.477	.463	.276	.253/.136/.136	1.59/1.21/1.23
CoOp + LP	bioclip_b16	mdca	.595	.415	.424	.156	.433/.156/.157	2.63/1.19/1.20
CoOp + LP	siglip2_b32	ce	.539	.487	.432	.459	.068/.064/.061	1.00/.998/.995
CoOp + LP	siglip2_b16	mbls	.510	.419	.369	.417	.051/.075/.073	1.04/1.04/1.04
CORAL	convnext_t	ce	.883	.676	.623	.349	.198/.136/.148	1.38/1.01/1.11
CORAL	dinov2_lora	ce	.853	.667	.625	.340	.143/.144/.150	1.05/1.05/1.11
ERM	dinov2_lora	ce	.857	.707	.646	.419	.154/.115/.119	1.12/.944/.965
Fine-tune	swin_t	mbls	.893	.655	.600	.298	.223/.171/.182	1.56/1.17/1.31
Fine-tune	convnext_t	ls	.888	.676	.611	.319	.144/.168/.177	.926/.982/1.04
Fine-tune	retfound_mae	mdca	.887	.525	.417	.291	.370/.276/.288	2.69/1.50/1.57
Fine-tune	retfound_dinov2	mbls	.886	.583	.501	.358	.285/.192/.207	1.93/1.19/1.33
Fine-tune	densenet121	ce	.882	.626	.581	.309	.238/.139/.158	1.88/1.13/1.30
Fine-tune	resnet50	mdca	.881	.666	.620	.319	.184/.110/.126	1.48/1.05/1.17
Fine-tune	inception_v3	mbls	.877	.664	.624	.405	.200/.139/.147	1.25/.956/1.05
Fine-tune	vgg16	mbls	.875	.671	.625	.371	.169/.135/.147	1.28/1.10/1.24
Fine-tune	deit_b16	mdca	.868	.622	.560	.313	.230/.158/.167	1.88/1.28/1.42
Fine-tune	vit_b16_in21k	mdca	.863	.596	.535	.336	.251/.180/.190	2.02/1.31/1.42
Fine-tune	efficientnet_b0	ce	.861	.651	.596	.266	.188/.108/.122	1.34/.974/1.04
Fine-tune	vit_b16_mae	ce	.856	.587	.532	.302	.272/.196/.205	2.17/1.42/1.59
Fine-tune	mobilenetv3_l	ls	.846	.648	.596	.238	.082/.130/.144	.855/.923/.993
Fine-tune, NLL rule	convnext_t	ce	.870	.677	.619	.324	.185/.133/.147	1.35/1.04/1.16
Fine-tune, NLL rule	resnet50	ce	.857	.647	.601	.254	.192/.117/.133	1.57/1.12/1.28
Group DRO	convnext_t	ce	.849	.707	.649	.417	.082/.091/.101	.886/.926/1.04
Group DRO	dinov2_lora	ce	.798	.733	.677	.486	.050/.035/.034	.732/.707/.732
Linear probe	medsiglip	class/ls	.920	.728	.654	.347	.024/.079/.093	.647/.694/.739
Linear probe	flair†	class/focal	.900	.721	.693	.703	.088/.117/.117	.705/.777/.775
Linear probe	dinov2_b14	class/mbls	.889	.674	.616	.434	.104/.103/.115	.843/.842/.875
Linear probe	retfound_mae	class/ls	.886	.455	.351	.243	.368/.412/.414	1.43/1.71/1.69
Linear probe	retfound_dinov2	class/ce	.868	.606	.522	.115	.185/.147/.158	1.16/1.03/1.08
Linear probe	siglip2_l16	class/mbls	.866	.605	.562	.206	.167/.159/.170	1.11/1.08/1.16
Linear probe	vit_b16_mae	class/ls	.866	.579	.536	.102	.136/.190/.200	.924/1.02/1.03
Linear probe	vit_b16_in21k	class/ls	.861	.608	.572	.292	.119/.170/.182	.983/1.09/1.15
Linear probe	metaclip_b16	class/ls	.861	.599	.551	.101	.114/.177/.188	.908/1.01/1.05
Linear probe	siglip2_b16	class/ls	.860	.546	.520	.484	.130/.187/.201	1.04/1.15/1.22
Linear probe	siglip_b16	class/ls	.856	.567	.540	.169	.130/.189/.201	.988/1.10/1.13
Linear probe	bioclip2_l14	class/ls	.853	.538	.520	.365	.164/.212/.227	1.05/1.15/1.17
Linear probe	densenet121	class/mdca	.852	.569	.522	.121	.228/.203/.211	1.38/1.26/1.35
Linear probe	siglip_l16	class/ce	.851	.524	.508	.352	.237/.235/.246	1.46/1.45/1.57
Linear probe	swin_t	class/mbls	.850	.584	.523	.176	.198/.180/.191	1.20/1.14/1.17
Linear probe	convnext_t	class/mbls	.848	.615	.553	.218	.145/.137/.147	.999/.981/1.01
Linear probe	clip_b16	class/ls	.844	.595	.528	.477	.104/.162/.171	.950/1.07/1.14
Linear probe	deit_b16	class/mdca	.844	.595	.537	.272	.179/.155/.167	1.11/1.05/1.09
Linear probe	bioclip_b16	class/mbls	.830	.539	.479	.164	.207/.198/.212	1.22/1.19/1.23
Linear probe	mobilenetv3_l	class/ls	.825	.583	.525	.137	.089/.153/.165	.968/1.07/1.11

Table S6: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

Setting	Encoder	Config	Val	Test	Eq-src	Chákṣu	ECE R/T/F	NLL R/T/F
Linear probe	siglip2_b32	class/ls	.824	.554	.517	.461	.130/.177/.188	1.01/1.10/1.13
Linear probe	efficientnet_b0	class/ce	.824	.598	.534	.196	.210/.179/.185	1.32/1.19/1.18
Linear probe	resnet50	class/mbls	.815	.650	.576	.334	.081/.056/.060	.888/.858/.868
Linear probe	biomedclip	class/mdca	.803	.479	.459	.510	.274/.271/.281	1.34/1.33/1.36
Linear probe	inception_v3	class/mbls	.799	.543	.500	.347	.220/.190/.198	1.22/1.14/1.15
Linear probe	vgg16	class/mbls	.793	.534	.480	.267	.234/.157/.179	1.89/1.56/1.36
MLP probe	dinov2_b14	none/ce	.870	.598	.524	.318	.263/.172/.183	2.31/1.26/1.42
MLP probe	siglip2_l16	none/ce	.859	.559	.481	.140	.281/.194/.194	2.92/1.73/1.73
MLP probe	metaclip_b16	none/ce	.853	.524	.473	.122	.360/.260/.269	3.04/1.53/1.69
MLP probe	siglip_b16	none/ce	.851	.570	.542	.330	.279/.182/.189	2.71/1.42/1.57
MLP probe	siglip_l16	none/ce	.847	.546	.469	.355	.346/.212/.216	4.72/1.86/2.02
MLP probe	siglip2_b16	none/ce	.845	.542	.510	.431	.292/.205/.217	2.95/1.71/2.05
MLP probe	bioclip2_l14	none/ce	.838	.514	.457	.318	.349/.234/.245	3.01/1.53/1.66
MLP probe	clip_b16	none/ce	.837	.545	.487	.450	.344/.250/.256	4.01/1.94/2.18
MLP probe	bioclip_b16	none/ce	.827	.502	.434	.089	.328/.252/.255	2.14/1.50/1.56
MLP probe	siglip2_b32	none/ce	.817	.546	.503	.490	.302/.181/.189	3.26/1.52/1.74
MLP probe	biomedclip	none/ce	.806	.525	.506	.475	.272/.185/.194	2.84/1.85/2.08
SWAD	convnext_t	ce	.883	.672	.622	.309	.208/.139/.149	1.57/1.06/1.16
SWAD	dinov2_lora	ce	.832	.716	.655	.436	.127/.092/.097	.964/.854/.879
Tip-Adapter	biomedclip	–	.597	.478	.491	.617	.332/.091/.096	1.75/1.02/1.03
Tip-Adapter	siglip2_l16	–	.550	.497	.534	.533	.140/.128/.117	1.00/.930/.914
Tip-Adapter	clip_b16	–	.473	.456	.466	.424	.082/.071/.077	1.04/1.08/1.07
Tip-Adapter (acc. rule)	biomedclip	–	.597	.478	.491	.617	.332/.091/.096	1.75/1.02/1.03
Tip-Adapter (acc. rule)	siglip2_l16	–	.550	.497	.534	.533	.140/.128/.117	1.00/.930/.914
Tip-Adapter (acc. rule)	clip_b16	–	.473	.456	.466	.424	.082/.071/.077	1.04/1.08/1.07
Tip-Adapter-F	biomedclip	–	.666	.495	.453	.532	.225/.147/.152	1.28/1.10/1.11
Tip-Adapter-F	siglip2_l16	–	.602	.538	.551	.528	.086/.157/.155	1.09/.986/.969
Tip-Adapter-F	clip_b16	–	.564	.508	.511	.459	.109/.095/.090	1.05/.997/.995
Tip-Adapter-F (acc. rule)	biomedclip	–	.666	.495	.453	.532	.225/.147/.152	1.28/1.10/1.11
Tip-Adapter-F (acc. rule)	siglip2_l16	–	.602	.538	.551	.528	.086/.157/.155	1.09/.986/.969
Tip-Adapter-F (acc. rule)	clip_b16	–	.564	.508	.511	.459	.109/.095/.090	1.05/.997/.995
WiSE-FT	siglip2_l16	fully-tuned	.886	.634	.587	.406	.236/.168/.179	1.84/1.24/1.38
Zero-shot	flair†	descriptive_single	.653	.712	.624	.567	.162/.154/.137	1.09/.867/.827
Zero-shot	medsiglip	descriptive_single	.650	.709	.606	.615	.207/.032/.032	.838/.690/.690
Zero-shot	biomedclip	descriptive_single	.503	.471	.535	.562	.229/.061/.074	1.29/1.00/1.02
Zero-shot	metaclip_b16	descriptive_single	.476	.403	.406	.374	.093/.119/.119	1.06/1.07/1.07
Zero-shot	siglip2_b32	descriptive_single	.473	.463	.501	.505	.028/.069/.063	1.05/1.07/1.07
Zero-shot	siglip2_l16	clip_ensemble	.461	.481	.494	.374	.031/.070/.085	1.06/1.17/1.13
Zero-shot	siglip_b16	cupl	.437	.433	.508	.503	.068/.090/.092	.959/.975/.973
Zero-shot	siglip2_b16	descriptive_single	.418	.509	.539	.518	.107/.176/.167	1.06/1.07/1.06
Zero-shot	bioclip_b16	descriptive_single	.366	.457	.518	.525	.358/.182/.183	3.79/1.10/1.10
Zero-shot	siglip_l16	cupl	.364	.357	.453	.500	.280/.162/.165	1.24/1.09/1.10
Zero-shot	bioclip2_l14	clip_ensemble	.344	.362	.281	.074	.326/.180/.186	1.54/1.16/1.18
Zero-shot	clip_b16	clip_ensemble	.327	.394	.445	.173	.055/.096/.094	1.10/1.15/1.16
Multiple choice	Fundus-R1-7B	avg6	–	.576	.569	.463	–	–
Multiple choice	Qwen2.5-VL-7B-Instruct	avg6	–	.424	.462	.488	–	–
Multiple choice	llava-med-v1.5-mistral-7b-hf	avg6	–	.347	.448	.507	–	–
Multiple choice	llava-onevision-qwen2-7b-ov-hf	avg6	–	.407	.327	.096	–	–

Reading. Within each setting the encoder ranking on test differs from that on validation (Sect. S5). MedSigLIP is the strongest frozen encoder across probe settings, and FLAIR is comparable only on sources it did not see in pretraining. Fine-tuned RETFound reaches high validation (0.886–0.887) but 0.525–0.583 on test, despite fundus pretraining. TS reduces ECE and NLL for most fine-tuned models but leaves ECE far

Table S4. Training recipes. Every setting selects by validation bAcc unless stated; post-hoc calibration always uses Cal-A, thresholds Cal-B.

Fine-tuning	Native input size and normalization; AdamW, lr 10^{-4} (head 10^{-3}), wd 0.05, cosine with one warm-up epoch, 15 epochs, batch 32; (source, class)-balanced sampling; crop, flip, rotation and colour augmentation; mixed precision; five losses (CE; label smoothing 0.1; focal γ=3; MDCA β=1; MbLS m=10, λ=0.1); 3 seeds. NLL-rule arm: checkpoint by validation NLL.
Linear, MLP probes	Frozen embeddings standardized on train; AdamW lr 10^{-3}, wd 10^{-4}, 40 epochs, batch 256; weighting none, class or (class, source); MLP: one hidden layer of 512.
16-shot probe	16 images per class, one per leakage group, redrawn per seed, 300 epochs.
Zero-shot	Three prompt banks (Table S5); bank chosen by validation bAcc.
CoOp	16 learnable context tokens, frozen towers, SGD lr 0.002, cosine, 200 full-batch steps on the 16 shots; ctx-init: 4 tokens from "a photo of a"; CoOp+LP: then a linear head, 200 steps, lr 10^{-3}; loss variants CE, focal, LS, MbLS, MDCA.
Tip-Adapter	Official recipe on the same shots; (α, β) grid and bank by validation bAcc (official accuracy rule kept as sensitivity); -F: keys fine-tuned 20 epochs, AdamW lr 10^{-3}.
ERM, Group DRO, CORAL	Fine-tuning recipe with ConvNeXt-T, or DINOv2-B/14 with LoRA rank 8 on query and value; Group DRO η=0.01, group = development source; CORAL λ=1.
SWAD	Dense averaging over segments of 150 updates, official loss-valley rule (3, 6, 0.3); replaces the validation checkpoint.
WiSE-FT	Head from the zero-shot classifier, AdamW lr 10^{-5}, wd 0.1, 15 epochs; interpolation $\alpha \in \{0, 0.1, \ldots, 1\}$ chosen by validation bAcc.
Multiple choice	Image plus a three-option question; next-token log-likelihoods of the option letters averaged over all six option orders.
LANTERN	Table S35.

above that of frozen probes. For transparency, the best frozen probe judged by test itself, a class-weighted cross-entropy MedSigLIP linear probe, reached test bAcc 0.778, equal-source 0.702, raw ECE 0.065 and recall G 0.657; validation selected the label-smoothed variant instead.

S4 Per-Source and Per-Class Results

Table S7: bAcc per unseen source (raw), equal-source mean and worst test source, for LANTERN and the validation-selected representatives.

Method	Paraguay	HYGD	FIVES	PAPILA	PALM	HRF	AFIO	Chákṣu	Eq-src	Worst
LANTERN	.900	.818	.735	.706	.700	.622	.584	.731	.724	.584
16-shot probe, medsiglip	.874	.756	.668	.499	.723	.711	.548	.675	.683	.499
CoOp, siglip_b16	.728	.130	.489	.197	.768	.415	.476	.318	.458	.130
CoOp ctx-init, biomedclip	.753	.038	.481	.435	.563	.400	.456	.384	.447	.038
CoOp + LP, siglip_b16	.735	.131	.496	.197	.781	.430	.471	.314	.463	.131
CORAL, convnext_t	.929	.263	.688	.413	.911	.622	.533	.349	.623	.263
ERM, dinov2_lora	.884	.505	.596	.498	.816	.607	.614	.419	.646	.498
Fine-tune, swin_t	.913	.237	.647	.468	.836	.600	.501	.298	.600	.237
Fine-tune, NLL rule, convnext_t	.917	.283	.704	.449	.880	.585	.516	.324	.619	.283
Group DRO, convnext_t	.920	.316	.704	.592	.870	.600	.539	.417	.649	.316
Linear probe, medsiglip	.934	.477	.559	.493	.907	.637	.571	.347	.654	.477
MLP probe, dinov2_b14	.889	.092	.631	.285	.747	.533	.492	.318	.524	.092
SWAD, convnext_t	.928	.243	.709	.409	.901	.615	.550	.309	.622	.243
Tip-Adapter, biomedclip	.399	.744	.427	.496	.480	.400	.493	.617	.491	.399
Tip-Adapter-F, biomedclip	.624	.283	.465	.471	.472	.407	.448	.532	.453	.283
WiSE-FT, siglip2_l16	.881	.250	.626	.423	.801	.600	.529	.406	.587	.250
Zero-shot, medsiglip	.729	.719	.723	.622	.379	.556	.512	.615	.606	.379

Table S8: Class recalls on the multi-class test sources.

Method	FIVES			AFIO			PAPILA		Paraguay	
	N	D	G	N	D	G	N	G	N	D
LANTERN	.687	.742	.777	.323	.778	.652	.819	.594	.847	.953
16-shot probe, medsiglip	.418	.757	.831	.075	.796	.773	.139	.858	.797	.951
CoOp, siglip_b16	.477	.717	.275	.164	1.00	.265	.110	.284	.624	.833
CoOp ctx-init, biomedclip	.522	.840	.083	.373	.926	.068	.349	.521	.653	.852
CoOp + LP, siglip_b16	.530	.723	.233	.144	.981	.288	.121	.272	.625	.845
CORAL, convnext_t	.440	.700	.924	.687	.685	.227	.405	.421	.920	.937
ERM, dinov2_lora	.171	.683	.933	.443	.815	.583	.413	.582	.806	.963
Fine-tune, swin_t	.328	.702	.910	.483	.907	.114	.318	.617	.872	.954
Fine-tune, NLL rule, convnext_t	.525	.710	.877	.597	.722	.227	.395	.502	.896	.937
Group DRO, convnext_t	.560	.695	.858	.458	.833	.326	.748	.437	.917	.922
Linear probe, medsiglip	.169	.692	.817	.647	.944	.121	.891	.096	.917	.951
MLP probe, dinov2_b14	.380	.650	.864	.701	.722	.053	.451	.119	.879	.898
SWAD, convnext_t	.520	.715	.893	.657	.759	.235	.362	.456	.916	.940
Tip-Adapter, biomedclip	.245	.362	.675	.353	.574	.553	.015	.977	.392	.406
Tip-Adapter-F, biomedclip	.180	.670	.544	.149	.870	.326	.030	.912	.443	.805
WiSE-FT, siglip2_l16	.323	.675	.881	.537	.852	.197	.228	.617	.834	.928
Zero-shot, medsiglip	.679	.800	.689	.254	.556	.727	.727	.517	.586	.871

Table S9: ECE after TS per source, and the equal-source mean.

Method	Paraguay	HYGD	FIVES	PAPILA	PALM	HRF	AFIO	Chákṣu	Eq-src
LANTERN	.038	.069	.048	.050	.109	.295	.256	.035	.123
16-shot probe, medsiglip	.075	.103	.075	.321	.124	.175	.301	.060	.168
CoOp, siglip_b16	.106	.607	.119	.446	.137	.216	.388	.199	.288
CoOp ctx-init, biomedclip	.090	.640	.167	.242	.153	.295	.356	.170	.278
CoOp + LP, siglip_b16	.103	.611	.116	.443	.139	.234	.412	.203	.294
CORAL, convnext_t	.055	.444	.184	.385	.181	.291	.249	.206	.256
ERM, dinov2_lora	.044	.215	.305	.288	.113	.278	.234	.113	.211
Fine-tune, swin_t	.043	.480	.250	.411	.170	.331	.333	.239	.288
Fine-tune, NLL rule, convnext_t	.051	.423	.171	.346	.175	.337	.280	.211	.255
Group DRO, convnext_t	.051	.405	.156	.100	.149	.263	.299	.081	.203
Linear probe, medsiglip	.032	.184	.249	.063	.107	.315	.285	.140	.176
MLP probe, dinov2_b14	.046	.617	.179	.328	.130	.303	.313	.215	.274
SWAD, convnext_t	.051	.466	.166	.390	.199	.288	.242	.219	.258
Tip-Adapter, biomedclip	.070	.315	.065	.362	.184	.137	.085	.170	.174
Tip-Adapter-F, biomedclip	.053	.336	.109	.568	.178	.317	.315	.165	.268
WiSE-FT, siglip2_l16	.063	.442	.250	.484	.083	.338	.294	.084	.279
Zero-shot, medsiglip	.064	.133	.052	.132	.270	.192	.209	.214	.150

Table S10. LANTERN ECE after TS per source on the published split, three seeds pooled, with 95% bootstrap intervals over leakage groups (1,000 draws).

Source	Images	Groups	ECE (TS)	95% CI
Paraguay	1,431	1,271	.038	[.029, .054]
HYGD	548	183	.069	[.054, .109]
FIVES	586	574	.048	[.037, .083]
PAPILA	420	210	.050	[.045, .091]
PALM	161	161	.109	[.088, .184]
HRF	45	45	.295	[.185, .432]
AFIO	129	65	.256	[.196, .365]
Chákṣu	1,343	1,306	.035	[.032, .061]

Reading. Paraguay is easy for every method, whereas AFIO and HRF are hard for all. D transfers well everywhere; the differences between methods come mostly from G and from N on G-heavy sources. Per-source ECE after a single global temperature remains high for every method on HRF and AFIO, small sets where a pooled ECE hides the error. Table S10 shows that their intervals are wide (HRF 0.185–0.432 with 45 images), so calibration on sets this small cannot be certified either way; a deployment site needs a local calibration set of a few hundred images (Table S14).

S5 Reliability

Table S11: Selective prediction on test with thresholds fitted on Cal-B, raw/TS/FTS. Cov 80, Cov 90: thresholds for that coverage on Cal-B. Risk 10: SGR bound of 10%.

		Cov 80				Cov 90		Risk 10		Chákṣu Cov 80	
Method		Cov.	Risk	G→N	D→N	Cov.	Risk	Cov.	Risk	Cov.	Risk
LANTERN	R	.713	.100	.059	.032	.857	.139	.665	.088	.354	.217
	T	.710	.094	.056	.031	.861	.139	.661	.082	.365	.218
	F	.710	.095	.058	.030	.860	.138	.662	.083	.372	.216
16-shot probe, medsiglip	R	.757	.196	.012	.017	.883	.245	.354	.060	.528	.307
	T	.753	.195	.012	.016	.876	.241	.353	.057	.509	.307
	F	.752	.195	.012	.016	.876	.241	.357	.058	.504	.307
CoOp, siglip_b16	R	.801	.477	.039	.086	.898	.489	.000	–	.598	.648
	T	.797	.476	.038	.089	.898	.488	.000	–	.572	.649
	F	.798	.476	.038	.086	.899	.489	.000	–	.586	.647
CoOp ctx-init, biomedclip	R	.761	.454	.200	.068	.869	.469	.000	–	.808	.395
	T	.745	.453	.187	.065	.854	.468	.000	–	.811	.398
	F	.749	.452	.189	.067	.857	.468	.000	–	.812	.396
CoOp + LP, siglip_b16	R	.793	.470	.040	.085	.895	.484	.000	–	.572	.673
	T	.796	.470	.040	.090	.898	.484	.000	–	.551	.675
	F	.797	.469	.039	.088	.898	.484	.000	–	.559	.674
CORAL, convnext_t	R	.668	.218	.157	.029	.807	.253	.727	.233	.530	.354
	T	.655	.212	.144	.028	.789	.245	.706	.224	.525	.353
	F	.668	.218	.158	.029	.807	.254	.726	.233	.529	.354
ERM, dinov2_lora	R	.726	.228	.075	.013	.848	.262	.697	.224	.566	.318
	T	.715	.223	.067	.012	.829	.255	.689	.221	.552	.314
	F	.723	.227	.073	.013	.839	.259	.695	.222	.561	.318
Fine-tune, swin_t	R	.708	.257	.156	.025	.819	.283	.766	.269	.457	.489
	T	.694	.250	.147	.024	.803	.276	.752	.263	.444	.484
	F	.709	.257	.156	.025	.819	.283	.766	.269	.458	.489
Fine-tune, NLL rule, convnext_t	R	.706	.219	.141	.033	.824	.253	.765	.237	.497	.423
	T	.690	.213	.126	.031	.809	.246	.755	.230	.489	.420
	F	.707	.220	.143	.033	.826	.254	.766	.238	.497	.423
Group DRO, convnext_t	R	.734	.180	.086	.042	.834	.208	.661	.162	.569	.249
	T	.738	.182	.089	.042	.838	.210	.664	.165	.570	.249
	F	.751	.187	.097	.044	.850	.215	.668	.167	.572	.249
Linear probe, medsiglip	R	.602	.127	.057	.022	.750	.161	.794	.176	.323	.271
	T	.610	.129	.061	.022	.765	.167	.800	.178	.308	.275
	F	.620	.132	.067	.023	.771	.169	.811	.183	.299	.276
MLP probe, dinov2_b14	R	.692	.290	.304	.048	.834	.335	.679	.288	.535	.461
	T	.672	.281	.281	.045	.812	.326	.666	.278	.524	.459
	F	.685	.287	.298	.047	.824	.330	.673	.285	.532	.462
SWAD, convnext_t	R	.708	.225	.172	.032	.832	.257	.820	.253	.478	.401
	T	.690	.216	.156	.030	.813	.246	.804	.245	.471	.399
	F	.704	.223	.167	.032	.825	.252	.816	.250	.475	.399
Tip-Adapter, biomedclip	R	.798	.549	.095	.153	.897	.554	.000	–	.794	.495
	T	.757	.547	.089	.113	.874	.549	.000	–	.930	.490
	F	.762	.546	.091	.116	.872	.548	.000	–	.923	.490
Tip-Adapter-F, biomedclip	R	.769	.526	.079	.045	.882	.526	.000	–	.810	.409
	T	.771	.524	.077	.046	.887	.525	.028	.446	.828	.416
	F	.770	.524	.077	.046	.887	.526	.030	.458	.828	.416
WiSE-FT, siglip2_l16	R	.659	.270	.100	.028	.804	.306	.679	.276	.430	.286
	T	.641	.260	.088	.025	.780	.295	.661	.268	.417	.280
	F	.654	.267	.097	.027	.798	.303	.673	.274	.426	.283
Zero-shot, medsiglip	R	.727	.230	.122	.052	.868	.270	.190	.040	.745	.581
	T	.711	.224	.107	.052	.861	.267	.182	.033	.689	.576
	F	.711	.224	.107	.052	.861	.267	.182	.033	.689	.576

Table S12: Conformal sets on test, quantiles from Cal-B. Coverage/mean size at nominal 0.90; coverage at nominal 0.95.

Method		LAC .90	APS .90	Class-LAC .90	LAC .95	APS .95	Class-LAC .95	Chákṣu LAC .90
LANTERN	R	.887/1.20	.935/1.73	.882/1.25	.955	.954	.942	.826/1.42
	T	.885/1.20	.955/1.89	.884/1.25	.958	.958	.944	.828/1.43
	F	.885/1.20	.922/1.70	.884/1.25	.954	.939	.941	.828/1.43
16-shot probe, medsiglip	R	.913/1.61	.923/1.99	.823/1.40	.966	.994	.919	.930/2.09
	T	.912/1.61	.922/1.98	.817/1.39	.967	.996	.916	.929/2.08
	F	.909/1.60	.925/2.01	.822/1.39	.966	.996	.922	.927/2.09
CoOp, siglip_b16	R	.761/1.80	.866/2.24	.761/1.83	.843	.947	.856	.817/2.26
	T	.760/1.80	.862/2.24	.760/1.83	.839	.944	.858	.811/2.23
	F	.760/1.79	.852/2.20	.773/1.85	.839	.986	.858	.802/2.18
CoOp ctx-init, biomedclip	R	.777/1.89	.762/1.91	.784/1.95	.846	.878	.845	.854/1.77
	T	.766/1.84	.760/1.90	.783/1.94	.835	.878	.846	.856/1.75
	F	.759/1.81	.750/1.87	.782/1.93	.825	.875	.850	.858/1.75
CoOp + LP, siglip_b16	R	.755/1.78	.809/2.01	.761/1.81	.839	.949	.847	.802/2.27
	T	.753/1.77	.808/2.00	.761/1.81	.837	.983	.849	.792/2.22
	F	.755/1.77	.795/1.96	.773/1.84	.832	.985	.853	.785/2.18
CORAL, convnext_t	R	.716/1.09	.815/1.68	.668/1.13	.780	.823	.754	.635/1.15
	T	.710/1.07	.807/1.60	.663/1.10	.773	.819	.749	.637/1.15
	F	.713/1.08	.796/1.56	.666/1.11	.769	.826	.741	.637/1.16
ERM, dinov2_lora	R	.728/1.10	.870/1.72	.697/1.13	.801	.885	.765	.684/1.17
	T	.722/1.08	.870/1.72	.689/1.11	.796	.884	.758	.679/1.15
	F	.723/1.08	.861/1.68	.692/1.11	.793	.882	.757	.679/1.15
Fine-tune, swin_t	R	.682/1.08	.794/1.65	.635/1.05	.737	.803	.720	.566/1.18
	T	.676/1.06	.786/1.59	.631/1.03	.732	.798	.715	.559/1.16
	F	.680/1.07	.784/1.67	.634/1.04	.730	.806	.714	.562/1.17
Fine-tune, NLL rule, convnext_t	R	.712/1.08	.829/1.73	.679/1.08	.771	.838	.769	.604/1.18
	T	.708/1.06	.828/1.71	.674/1.06	.764	.836	.762	.604/1.18
	F	.711/1.07	.806/1.58	.679/1.07	.763	.829	.757	.602/1.18
Group DRO, convnext_t	R	.767/1.13	.844/1.65	.733/1.09	.810	.854	.814	.787/1.29
	T	.769/1.13	.844/1.65	.733/1.09	.811	.854	.815	.788/1.29
	F	.771/1.14	.838/1.65	.735/1.10	.811	.855	.810	.790/1.30
Linear probe, medsiglip	R	.743/0.99	.883/1.68	.682/0.92	.857	.893	.765	.631/1.09
	T	.751/1.01	.888/1.72	.687/0.94	.863	.897	.765	.626/1.08
	F	.754/1.01	.855/1.55	.691/0.95	.857	.876	.760	.626/1.08
MLP probe, dinov2_b14	R	.658/1.11	.882/2.28	.637/1.13	.746	.931	.703	.576/1.17
	T	.651/1.09	.798/1.79	.629/1.11	.739	.807	.695	.573/1.16
	F	.652/1.09	.777/1.67	.633/1.12	.738	.798	.696	.573/1.17
SWAD, convnext_t	R	.698/1.04	.827/1.73	.668/1.05	.761	.834	.766	.562/1.09
	T	.690/1.01	.822/1.69	.662/1.02	.753	.831	.758	.558/1.08
	F	.693/1.02	.802/1.54	.666/1.03	.753	.825	.750	.560/1.09
Tip-Adapter, biomedclip	R	.947/2.50	.989/2.92	.785/2.00	.979	.989	.866	.994/2.27
	T	.970/2.54	.993/2.95	.756/1.85	.988	.993	.845	.000/2.24
	F	.961/2.52	.992/2.94	.764/1.90	.984	.992	.840	.997/2.25
Tip-Adapter-F, biomedclip	R	.819/2.09	.989/2.92	.764/1.88	.889	.989	.838	.897/2.09
	T	.825/2.09	.992/2.94	.766/1.86	.899	.992	.841	.901/2.10
	F	.817/2.08	.948/2.70	.763/1.87	.889	.990	.837	.898/2.10
WiSE-FT, siglip2_l16	R	.676/1.12	.805/1.78	.633/1.13	.738	.813	.707	.717/1.23
	T	.668/1.10	.804/1.77	.628/1.11	.732	.812	.700	.707/1.20
	F	.671/1.11	.790/1.70	.631/1.12	.729	.811	.701	.709/1.21
Zero-shot, medsiglip	R	.923/1.74	.896/1.80	.912/1.87	.981	.996	.967	.852/1.91
	T	.924/1.76	.920/1.89	.916/1.90	.982	.994	.968	.842/1.90
	F	.924/1.76	.920/1.89	.916/1.90	.982	.994	.968	.842/1.90

Reading. Methods whose sets stay near nominal coverage (16-shot, zero-shot) do so with mean set sizes of 1.6 to 1.8 out of three classes, which is of little use. Methods with compact sets lose coverage. TS keeps the ranking of confidence, so it barely changes coverage or risk at a fixed threshold.

Reading. On the other half of Cal-B, the promises hold almost exactly; on test and Chákṣu they fail. The failure therefore comes from the new sources, not from a small calibration set.

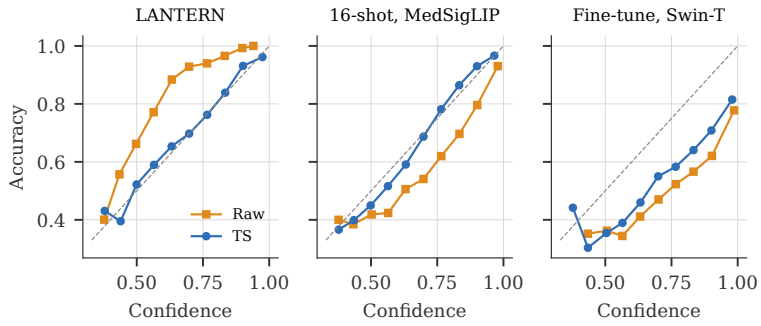


Fig. S1. Test reliability on the published split, seeds pooled, bins with at least 30 images. Raw LANTERN is underconfident and TS brings it near the diagonal; fine-tuned Swin-T stays overconfident after TS.

Table S14: Calibration budget: fitted T on all of Cal-A, the oracle T fitted on test, and test NLL after TS with natural random subsets of Cal-A (median of 100 draws).

Setting	Model	T	T oracle	NLL, 50	NLL, 200	NLL, 1000	NLL, all
16-shot probe	biomedclip	1.59	2.88	1.05	1.05	1.05	1.05
16-shot probe	clip_b16	1.80	3.28	1.08	1.08	1.07	1.07
16-shot probe	dinov2_b14	2.80	4.47	.997	.994	.993	.993
16-shot probe	resnet50	3.75	2.61	.979	.976	.977	.977
16-shot probe	siglip2_l16	1.57	1.75	.929	.926	.926	.926
CoOp	biomedclip	2.07	6.48	1.23	1.22	1.23	1.23
CoOp	clip_b16	1.43	4.17	1.21	1.20	1.19	1.20
CoOp	siglip2_l16	.780	.893	1.07	1.07	1.07	1.07
CoOp ctx-init	biomedclip	2.22	6.34	1.19	1.17	1.18	1.18
CoOp ctx-init	clip_b16	1.25	3.88	1.20	1.19	1.19	1.19
CoOp ctx-init	siglip2_l16	1.32	3.97	1.20	1.20	1.19	1.19
CoOp + LP	biomedclip	2.17	6.89	1.24	1.23	1.24	1.24
CoOp + LP	clip_b16	1.51	4.39	1.21	1.20	1.19	1.19
CoOp + LP	siglip2_l16	.358	.385	.898	.896	.896	.895
Fine-tune	resnet50	1.52	3.65	1.02	1.01	1.01	1.01
Fine-tune, NLL rule	resnet50	1.69	4.68	1.13	1.11	1.12	1.12
Linear probe	biomedclip	.866	1.79	1.06	1.04	1.04	1.04
Linear probe	clip_b16	.889	1.83	.971	.944	.951	.950
Linear probe	dinov2_b14	.999	2.04	.926	.891	.904	.901
Linear probe	resnet50	1.16	1.87	.924	.911	.911	.910
Linear probe	siglip2_l16	.999	2.22	1.01	.989	.993	.993
Multiple choice	Fundus-R1-7B	1.87	2.24	.821	.816	.817	.817
Multiple choice	Qwen2.5-VL-7B-Instruct	1.78	3.39	1.07	1.08	1.08	1.08
Multiple choice	llava-med-v1.5-mistral-7b-hf	.725	.483	1.07	1.07	1.07	1.07
Multiple choice	llava-onevision-qwen2-7b-ov-hf	1.96	11.32	1.30	1.28	1.28	1.28
MLP probe	biomedclip	1.74	13.47	1.90	1.85	1.85	1.85
MLP probe	clip_b16	2.36	16.03	1.99	1.97	1.95	1.94
MLP probe	dinov2_b14	2.09	7.61	1.32	1.25	1.26	1.26
MLP probe	siglip2_l16	1.91	11.20	1.78	1.75	1.74	1.73
Zero-shot	biomedclip	1000.00	137.44	1.10	1.10	1.10	1.10
Zero-shot	clip_b16	.785	2.12	1.15	1.15	1.15	1.15
Zero-shot	siglip2_l16	.604	1.69	1.17	1.18	1.17	1.17

Reading. About 200 calibration images give nearly the NLL of the full set, yet the fitted T stays far from the test oracle for several models: more calibration data from the same sources does not close the shift.

Table S13. Is the failure caused by too few calibration images? Cal-B is split by leakage group 100 times; thresholds fit on one half are read on the other half (same sources), on test and on Chákṣu (TS).

	Coverage at Cov 80			Risk at Risk 10			LAC .90 coverage		
Method	Other half	Test	Chákṣu	Other half	Test	Chákṣu	Other half	Test	Chákṣu
16-shot probe, medsiglip	.799	.754	.507	.054	.041	.132	.900	.911	.926
CoOp, siglip_b16	.799	.797	.573	–	–	–	.901	.760	.811
CoOp ctx-init, biomedclip	.800	.744	.810	.134	.257	.294	.900	.767	.856
CoOp + LP, siglip_b16	.799	.795	.548	–	–	–	.901	.756	.796
CORAL, convnext_t	.797	.650	.522	.068	.209	.357	.901	.710	.632
ERM, dinov2_lora	.799	.714	.550	.068	.210	.289	.900	.722	.682
Fine-tune, swin_t	.800	.695	.446	.069	.251	.487	.901	.675	.554
Fine-tune, NLL rule, convnext_t	.798	.688	.487	.068	.213	.424	.902	.709	.605
Group DRO, convnext_t	.797	.737	.569	.064	.150	.248	.900	.771	.789
Linear probe, medsiglip	.800	.610	.308	.070	.161	.318	.901	.752	.632
MLP probe, dinov2_b14	.801	.673	.526	.069	.258	.452	.901	.651	.573
SWAD, convnext_t	.798	.689	.470	.070	.230	.423	.901	.691	.560
Tip-Adapter, biomedclip	.797	.757	.928	–	–	–	.900	.970	.000
Tip-Adapter-F, biomedclip	.801	.772	.829	.084	.436	.642	.900	.827	.904
WiSE-FT, siglip2_l16	.799	.641	.420	.068	.258	.281	.900	.668	.708
Zero-shot, medsiglip	.799	.717	.696	.045	.027	.432	.902	.925	.844

Table S15. Trivial predictors, the floor for every metric.

View	Predictor	Acc.	bAcc	Eq-src	ECE	NLL
Test	uniform	.445	.333	.429	.112	1.10
Test	train_prior	.445	.333	.429	.063	1.30
Test	always_N	.445	.333	.429	–	–
Test	always_D	.288	.333	.214	–	–
Test	always_G	.267	.333	.357	–	–
Chákṣu	uniform	.861	.500	.500	.527	1.10
Chákṣu	train_prior	.861	.500	.500	.353	.978
Chákṣu	always_N	.861	.500	.500	–	–
Chákṣu	always_D	.000	.000	.000	–	–
Chákṣu	always_G	.139	.500	.500	–	–

Reading. A class-prior predictor reaches ECE 0.063 with chance bAcc, so low ECE alone says nothing about usefulness. The high overall rank correlation (0.781 here, over 847 rows that also count the accuracy-rule Tip-Adapter variants; 0.786 over the 816 configurations of the main paper) mostly reflects differences between families; within fine-tuning it is 0.053.

S6 Robustness Checks

Reading. Validation exceeded test in 98% of probe rows, risk exceeded its internal estimate in 89%, and LAC fell below its internal coverage in 84%; for LANTERN the shares were 94%, 88% and 94%. Ranking is not stable either: the 16-shot probe, best on the published split among representatives, is below full training on every encoder over the rotations.

Table S16. Spearman correlation between validation and test bAcc across configurations within each setting.

Setting	Configurations	Spearman
16-shot probe	130	.374
CoOp	70	.800
CoOp ctx-init	70	.927
CoOp + LP	50	.393
Fine-tune	65	.053
Linear probe	390	.344
MLP probe	11	.527
Tip-Adapter	3	.500
Tip-Adapter (acc. rule)	3	.500
Tip-Adapter-F	3	−.500
Tip-Adapter-F (acc. rule)	3	−.500
WiSE-FT	4	-1.00
Zero-shot	36	.831
All configurations	847	.781

Table S17. Fifty random 7/7 source rotations (published split excluded). Probes use class-weighted cross-entropy; arrows go from internal estimates on held-back Cal-B to unseen test sources, after TS.

Method	Val	Test	5–95%	Eq-src	Rec. G	ECE raw	ECE TS	Risk at Cov 80	LAC .90	Coverage .80
LANTERN	.853	.771	.706–.827	.747	.667	.146	.033	.080→.116	.898→.853	.796→.742
medsiglip, full	.922	.745	.621–.827	.725	.637	.095	.084	.042→.134	.901→.769	.795→.668
medsiglip, 16-shot	.814	.716	.626–.817	.713	.723	.123	.056	.164→.217	.899→.868	.803→.778
dinov2_b14, full	.892	.655	.529–.765	.650	.531	.123	.118	.077→.221	.899→.735	.794→.659
dinov2_b14, 16-shot	.713	.552	.432–.659	.571	.505	.192	.073	.296→.383	.900→.860	.801→.779
siglip2_l16, full	.876	.631	.549–.731	.617	.549	.155	.141	.102→.280	.900→.728	.804→.680
siglip2_l16, 16-shot	.722	.578	.476–.683	.562	.586	.160	.080	.292→.372	.901→.871	.796→.751
convnext_t, full	.860	.593	.483–.700	.609	.426	.149	.130	.112→.287	.901→.730	.800→.684
convnext_t, 16-shot	.679	.494	.379–.604	.504	.456	.223	.091	.352→.463	.900→.840	.795→.771

Reading. Text fusion is the component that matters most, in both accuracy and calibration. The bag is neutral, and between-site projection lowers accuracy across rotations although it helped on development data (Sect. S8).

Reading. Seed variation is an order of magnitude smaller than variation between source partitions, so single-seed rotations measure the partition effect. The projection-free variant stays ahead with seeds averaged.

Reading. The between-site directions hold 79–87% of every class-mean difference, about four times what a random subspace holds. Sites differ in severity and case mix within each label, so their within-class differences are themselves disease directions; projecting them out removes signal, which explains the accuracy loss over the rotations.

Reading. Glaucoma recall grows with the number of glaucoma sources on the development side, so the weakness is one of site diversity, which more sources can address. LANTERN's advantage over the linear probe is largest when diversity is lowest (two sources: bAcc 0.761 against 0.678).

Table S18. LANTERN ablation over the 50 rotations (TS); standard deviation of bAcc across rotations.

Configuration	bAcc	SD	Eq-src	Rec. N	Rec. D	Rec. G	False G	ECE	Chákṣu
LANTERN	.771	.040	.747	.796	.850	.667	.081	.033	.666
without projection	.788	.041	.766	.813	.845	.708	.071	.034	.692
without bag	.768	.040	.745	.795	.852	.657	.080	.033	.663
without text	.754	.047	.728	.776	.827	.658	.100	.056	.626
single head only	.770	.055	.740	.792	.818	.701	.088	.053	.593
text only	.713	.036	.692	.693	.831	.614	.131	.022	.594

Table S19. Seed variability over the rotations: three training seeds per rotation (rotations where all three finished). Seed SD: standard deviation across seeds within a rotation, averaged; rotation SD: standard deviation of seed-averaged bAcc across rotations.

Configuration	Rotations	bAcc	Seed SD	Rotation SD	ECE	Recall G
LANTERN	50	.771	.003	.039	.033	.666
without projection	50	.789	.004	.039	.034	.709
text only	50	.713	.000	.036	.022	.614
Without projection minus LANTERN, seed-averaged: +0.018, higher on 45 of 50 rotations						

Reading. N–D and D–G stay near 0.90–0.95 while N–G is 0.64–0.75, so glaucoma is a ranking failure, not a threshold failure. Within-source learning recovers much of it (Chákṣu gap +0.372), so the labels are learnable and the loss is tied to the shift. AFIO stays low even within source.

Reading. More sources usually help on test (MedSigLIP 0.594 to 0.817 from one to seven sources) but not on Chákṣu (0.534 to 0.350), and more labels from the same sources give little after about 64 per class. Shift is not solved by more of the same data.

S7 Attempted Fixes and Other Probes

Reading. No label-free fix helps reliably. TENT changes little, BatchNorm adaptation harms ResNet-50 (notably on single-class PALM), and T3A is mostly negative. EM prior estimation is unstable and usually worsens ECE. True per-source priors raise bAcc to 0.81–0.86 for most settings, so class balance is a large part of the shift, but this oracle uses test labels, and single-class sources make it easy.

Table S28: K labelled images from each target source, read on the rest of that source: ECE with the frozen T against T re-fitted on K; LAC .90 coverage with the frozen quantile against a quantile from K, with the 5th percentile over draws.

Setting	Model	K	ECE frozen	ECE T from K	LAC frozen	LAC from K	LAC 5%
16-shot probe	medsiglip	10	.156	.128	.879	.897	.714
		25	.152	.093	.869	.920	.801
		50	.154	.087	.869	.899	.808
		100	.129	.067	.900	.901	.837
Linear probe	medsiglip	10	.172	.134	.677	.899	.704
		25	.153	.098	.682	.917	.785

Table S28: K labelled images from each target source, read on the rest of that source: ECE with the frozen T against T re-fitted on K; LAC .90 coverage with the frozen quantile against a quantile from K, with the 5th percentile over draws.

Setting	Model	K	ECE frozen	ECE T from K	LAC frozen	LAC from K	LAC 5%
		50	.154	.086	.682	.898	.797
		100	.134	.073	.672	.901	.841
Zero-shot	medsiglip	10	.163	.136	.884	.901	.722
		25	.155	.103	.884	.920	.809
		50	.156	.094	.884	.900	.813
		100	.121	.080	.919	.902	.843
ERM	dinov2_lora	10	.200	.140	.658	.904	.714
		25	.189	.110	.663	.920	.819
		50	.191	.104	.663	.899	.805
		100	.194	.089	.643	.901	.831
Group DRO	convnext_t	10	.189	.140	.712	.903	.715
		25	.178	.117	.721	.916	.803
		50	.180	.109	.721	.899	.800
		100	.160	.101	.718	.901	.837
Fine-tune	swin_t	10	.283	.125	.590	.904	.712
		25	.276	.095	.588	.921	.809
		50	.277	.088	.588	.901	.811
		100	.285	.066	.557	.902	.838

Reading. Ten labels from the new source already bring LAC coverage near 0.90 on average and re-fitted T lowers ECE, but the worst draws stay far below nominal at small K; this is a deployment recipe, not a guarantee.

Reading. CLAHE does not help fine-tuning or the 16-shot probe. The MedSigLIP linear probe rises on test, but its validation falls, so validation-based selection would not adopt it.

Reading. MSP accepts a third to two thirds of out-of-scope images. KNN rejects most of them and lowers risk, but largely by dropping whole hard sources; it is useful for flagging strange images rather than as a remedy for shift.

Reading. Generative multimodal models rarely call glaucoma. Mild DR (grade 1) is the hardest D grade for most models. Strong blur and compression mainly remove glaucoma recall, consistent with the disc being a small, detail-dependent region.

S8 LANTERN: Specification and Development

S8.1 Design search on development sources

All results in this section are leave-one-development-source-out on development data, equal-site mean, three seeds unless noted, and use no test row. Table S36 compares losses (cross-entropy; a conditional-logit pair loss across sites, "cross"; the same within sites, "strata"), a single head against a leave-one-source-out bag, the between-site projection ("nap"), and text fusion with one bank or all three, on pooled MedSigLIP features.

Table S36: Factorial comparison on pooled features (development, three seeds).

Arm	bAcc	ECE	NLL	Recall G
cross bag nap+allbanks	.794	.156	.574	.636
strata bag nap+allbanks	.792	.151	.573	.625

Table S36: Factorial comparison on pooled features (development, three seeds).

Arm	bAcc	ECE	NLL	Recall G
ce bag nap+allbanks	.791	.155	.581	.639
strata bag+allbanks	.784	.148	.584	.618
cross bag nap+text	.784	.137	.577	.633
cross bag+allbanks	.782	.149	.594	.557
strata single nap+allbanks	.782	.144	.578	.692
strata bag nap+text	.781	.131	.577	.630
ce bag nap+text	.780	.134	.584	.640
ce bag+allbanks	.777	.149	.604	.576
cross single nap+allbanks	.776	.141	.589	.574
strata bag+text	.774	.129	.589	.622
strata single nap+text	.773	.128	.583	.696
cross bag+text	.773	.134	.597	.570
cross single nap+text	.770	.125	.593	.589
strata single+allbanks	.770	.135	.595	.594
ce bag+text	.768	.132	.606	.589
ce single nap+allbanks	.765	.136	.601	.583
strata single+text	.762	.120	.599	.607
cross bag nap	.762	.133	.623	.625
cross single+allbanks	.760	.135	.616	.472
strata bag nap	.759	.127	.621	.603
ce single nap+text	.758	.117	.604	.585
cross single+text	.753	.123	.619	.485
ce single+allbanks	.752	.143	.622	.531
ce bag nap	.750	.129	.641	.629
strata bag	.747	.130	.640	.597
ce single+text	.746	.130	.624	.541
cross bag	.737	.146	.663	.481
strata single nap	.731	.107	.651	.712
cross single nap	.729	.126	.666	.526
strata single	.723	.129	.675	.565
ce bag	.721	.145	.687	.507
ce single nap	.704	.125	.694	.517
cross single	.691	.158	.715	.319
ce single	.675	.169	.741	.417

Reading. Every component adds a little alone; with all three in place the loss choice no longer matters (0.791–0.794). Text with all banks is the largest single step.

Table S37: Head input (pooled feature, last-layer token mean, three depths, both) and head recipe (base: 40 epochs, lr 10^{-3}, wd 10^{-4}, batch 256; reg: 12 epochs, lr 3×10^{-4}, wd 10^{-2}, batch 64), single head or bag with projection, with text.

Study	Arm	bAcc	ECE	NLL	Recall G	False G
recipe	last reg bag nap+text	.792	.151	.575	.764	.100
recipe	pooled base bag nap+text	.791	.155	.581	.639	.070
input	pooled bag nap+text	.791	.155	.581	.639	.070
recipe	last base bag nap+text	.787	.150	.582	.678	.078
input	last bag nap+text	.787	.150	.582	.678	.078
recipe	pooled reg bag nap+text	.787	.155	.580	.783	.109
input	pooled last bag nap+text	.782	.154	.594	.707	.093
recipe	pooled reg single+text	.759	.136	.621	.688	.109
recipe	last reg single+text	.759	.133	.616	.650	.098
recipe	last base single+text	.757	.132	.616	.547	.074
input	last single+text	.757	.132	.616	.547	.074
recipe	pooled base single+text	.752	.143	.622	.531	.071
input	pooled single+text	.752	.143	.622	.531	.071
input	pooled last single+text	.744	.149	.631	.546	.085
input	depths bag nap+text	.731	.118	.642	.613	.118
input	depths single+text	.696	.139	.678	.520	.126

Table S20. Why projection hurts. Share of the norm of each class-mean difference that lies in the removed between-site subspace (k directions), on the development data used by each head, against a random subspace of the same size; and alignment between the two glaucoma sites' difference and the N–G direction.

Source left out	k	N–D	N–G	D–G	N–G, random	| cos | G-site, N–G
none (all)	14	.868	.788	.811	.187	.098
Bangladesh Eye	11	.839	.548	.602	.225	–
OIA-DDR	12	.830	.839	.817	.208	.073
DeepDRiD	12	.850	.776	.762	.208	.083
GRAPE	13	.866	.819	.836	.147	–
IDRiD	12	.818	.784	.793	.209	.100
Jichi	12	.832	.726	.747	.218	.048
SUSTech-SYSU	12	.791	.785	.782	.207	.115

Table S21. Glaucoma recall over the 50 rotations by the number of development sources that contain glaucoma.

Method	G sources in development	Rotations	Recall G	Test bAcc
LANTERN	2	8	.554	.761
LANTERN	3	18	.672	.773
LANTERN	4	17	.700	.773
LANTERN	5	6	.690	.764
LANTERN	6	1	.762	.806
MedSigLIP LP	2	8	.469	.678
MedSigLIP LP	3	18	.628	.739
MedSigLIP LP	4	17	.667	.763
MedSigLIP LP	5	6	.747	.773
MedSigLIP LP	6	1	.959	.889

Reading. Heavier or learned modules repeatedly lost the glaucoma class at held-out sites, which led to the low-capacity design.

S8.2 Error analysis and disc-side bias

Reading. On OIA-DDR, 1,629 of 5,639 normal images were predicted G although the source holds no glaucoma. Between 0.9% and 13.7% of a site's images were wrong under all nine inputs; among normal IDRiD images, 28%. A global class offset did not fix the G confusion, so it is a ranking problem.

Table S42: Upgrades of LANTERN's input on development data: dimension (384 or 1152), mirrored training and averaging, NAP k, weight decay.

Arm	bAcc	NLL	Recall G	False G	N→G left	N→G right
d384 mirror k32 wd0.1	.792	.577	.658	.078	.127	.127
d384 mirror k16 wd0.1	.792	.577	.658	.078	.127	.127
d384 mirror k32 wd0.01	.792	.577	.652	.076	.125	.124
d384 mirror k16 wd0.01	.792	.577	.652	.076	.125	.124
d384 mirror k32 wd0.001	.792	.578	.653	.077	.124	.124
d384 mirror k16 wd0.001	.792	.578	.653	.077	.124	.124
d384 mirror k8 wd0.001	.790	.578	.650	.077	.126	.125

Table S42: Upgrades of LANTERN's input on development data: dimension (384 or 1152), mirrored training and averaging, NAP k, weight decay.

Arm	bAcc	NLL	Recall G	False G	N→G left	N→G right
d384 mirror k8 wd0.01	.790	.579	.648	.077	.126	.124
d384 mirror k8 wd0.1	.790	.578	.644	.077	.125	.124
d384 plain k32 wd0.01	.789	.576	.768	.103	.215	.141
d1152 mirror k32 wd0.01	.778	.603	.711	.100	.176	.165
d1152 plain k32 wd0.01	.770	.603	.823	.134	.296	.188

Reading. Mirroring removes the left-right gap and lowers false G but costs recall G; all 1152 dimensions are worse; k and weight decay hardly matter.

S8.3 Patch-level evidence

LANTERN's head input is the mean of patch tokens, and standardization, projection and the linear head are all linear, so each head's logits split exactly into per-patch contributions: $z_s(x) = \frac{1}{P}\sum_p W_s(I - U_s U_s^\top)x_p + b_s$. Figure S2 maps the glaucoma evidence (G logit minus the larger of N and D, averaged over heads) for the most and least confident glaucoma image of each glaucoma test source. Tokens recomputed for these images matched the stored ones to within 0.13 (feature scale about 90).

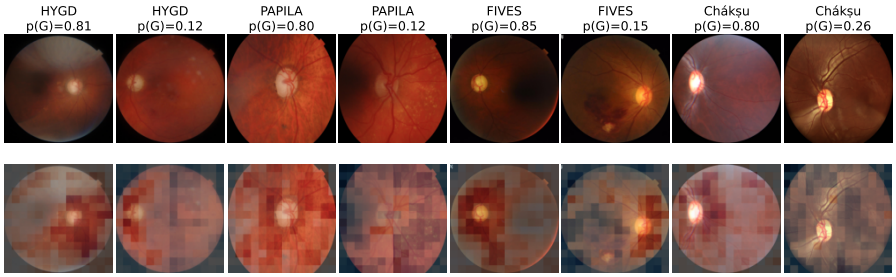


Fig. S2. Glaucoma evidence per patch (red: for G, blue: against), for the most and least confident glaucoma image of each source, $p(\mathrm{G})$ from raw fused logits. In confident cases the evidence concentrates on the optic disc; in missed cases it is weak and diffuse. Text scores are global and not mapped.

Reading. The maps are exact decompositions of the head, not post-hoc saliency, but they are not a validated localization of pathology.

S8.4 Later attempts

Two further components were tried with selection rules written before their results were read. *Label scope*: normal labels from DR-only sources were treated as the set {N, G}, with likelihood $p_\mathrm{N} + p_\mathrm{G}$. *Shift-aware reference*: temperature, thresholds and conformal

Table S22. Pairwise AUROC, mean over published test sources holding both classes, with bootstrap intervals over sources.

Pair	A	B	Sources	AUROC A	AUROC B	A−B	95% CI
D–G	16-shot probe	Fine-tune	3	.939	.916	+0.023	[-0.029, +0.077]
D–G	16-shot probe	Linear probe	3	.939	.917	+0.022	[-0.012, +0.059]
D–G	16-shot probe	Group DRO	3	.939	.942	-0.003	[-0.035, +0.033]
D–G	Linear probe	Fine-tune	3	.917	.916	+0.001	[-0.062, +0.081]
D–G	Group DRO	ERM	3	.942	.916	+0.025	[+0.010, +0.041]
N–D	16-shot probe	Fine-tune	4	.946	.900	+0.046	[+0.016, +0.074]
N–D	16-shot probe	Linear probe	4	.946	.928	+0.019	[-0.005, +0.053]
N–D	16-shot probe	Group DRO	4	.946	.923	+0.024	[+0.007, +0.045]
N–D	Linear probe	Fine-tune	4	.928	.900	+0.028	[-0.012, +0.069]
N–D	Group DRO	ERM	4	.923	.909	+0.013	[-0.021, +0.054]
N–G	16-shot probe	Fine-tune	4	.749	.643	+0.106	[+0.038, +0.203]
N–G	16-shot probe	Linear probe	4	.749	.678	+0.071	[-0.011, +0.152]
N–G	16-shot probe	Group DRO	4	.749	.704	+0.045	[-0.006, +0.118]
N–G	Linear probe	Fine-tune	4	.678	.643	+0.035	[-0.059, +0.135]
N–G	Group DRO	ERM	4	.704	.644	+0.060	[-0.015, +0.123]

Table S23. Trained on development sources against trained within the target (logistic regression on frozen features, mean over nine encoders; within-source by cross-validation). Gap: within minus development bAcc, with its bootstrap interval.

	bAcc		ECE		AUROC N–G		AUROC N–D		
Source	Dev	Within	Dev (TS)	Within	Dev	Within	Dev	Within	Gap
AFIO	.516	.635	.344	.153	.603	.745	.856	.829	+0.126 [-0.001, +0.249]
Chákṣu	.352	.718	.261	.044	.638	.806	–	–	+0.372 [+0.328, +0.415]
FIVES	.581	.857	.227	.049	.780	.961	.883	.969	+0.283 [+0.245, +0.320]
HRF	.583	.779	.295	.182	.477	.854	.926	.927	+0.193 [+0.053, +0.328]
PAPILA	.394	.757	.212	.070	.635	.832	–	–	+0.371 [+0.296, +0.445]
Paraguay	.817	.928	.046	.018	–	–	.934	.975	+0.115 [+0.095, +0.137]

quantiles were fitted on leave-one-source-out predictions rather than on Cal-B, with an optional normal certificate.

Table S45: Thresholds from Cal-B against the leave-one-source-out reference, 50 rotations, with and without the normal certificate.

Fitted on	Target	Test cov.	Test risk	FR	G→N	Chákṣu cov.	Chákṣu risk
Cal-B	cov. .80	.744	.117	.076	.097	.464	.444
Cal-B	cov. .90	.868	.153	.108	.160	.718	.463
Cal-B	risk .10	.641	.100	.060	.085	.357	.411
Cal-B + cert.	cov. .80	.601	.109	.035	.036	.338	.601
Cal-B + cert.	cov. .90	.661	.145	.037	.040	.475	.676
Cal-B + cert.	risk .10	.540	.094	.031	.034	.263	.528
LOSO reference	cov. .80	.830	.141	.096	.139	.639	.459
LOSO reference	cov. .90	.913	.167	.120	.186	.827	.469
LOSO reference	risk .10	.555	.074	.040	.042	.194	.402
LOSO reference + cert.	cov. .80	.650	.131	.034	.035	.432	.648
LOSO reference + cert.	cov. .90	.690	.158	.034	.035	.529	.696
LOSO reference + cert.	risk .10	.524	.072	.031	.031	.177	.436

Table S24. More training sources at a fixed size (300 images per class): test and Chákṣu bAcc for $K = 1, 2, 4, 7$ development sources (mean over random pools).

	Test				Chákṣu			
Encoder	1	2	4	7	1	2	4	7
clip_b16	.509	.478	.508	.558	.527	.446	.431	.494
convnext_t	.481	.514	.518	.557	.514	.491	.408	.440
dinov2_b14	.544	.618	.593	.720	.535	.555	.485	.543
flair†	.555	.633	.678	.758	.511	.624	.643	.679
medsiglip	.594	.725	.743	.817	.534	.561	.504	.350
resnet50	.547	.544	.503	.628	.564	.552	.362	.455
siglip2_l16	.512	.518	.548	.602	.494	.456	.387	.102
vit_b16_in21k	.417	.478	.542	.576	.504	.456	.347	.430

Table S25. Learning curve: change in test bAcc at k labels per class against the next smaller budget, with bootstrap intervals.

Encoder	k	ΔbAcc	95% CI
medsiglip	16	-0.008	[-0.161, +0.130]
medsiglip	64	+0.021	[-0.111, +0.124]
medsiglip	128	+0.040	[-0.060, +0.120]
medsiglip	512	+0.009	[-0.054, +0.078]
dinov2_b14	16	-0.130	[-0.257, +0.032]
dinov2_b14	64	-0.044	[-0.151, +0.088]
dinov2_b14	128	-0.016	[-0.114, +0.089]
dinov2_b14	512	+0.003	[-0.076, +0.060]
siglip2_l16	16	-0.050	[-0.129, +0.024]
siglip2_l16	64	-0.015	[-0.096, +0.087]
siglip2_l16	128	+0.025	[-0.080, +0.125]
siglip2_l16	512	+0.007	[-0.103, +0.056]

Reading. Label scope over-calls glaucoma at new sources. The shift-aware reference keeps coverage and risk promises far more often; risk at matched coverage is identical for both, so the arms differ only in where the threshold sits.

S9 Statistical Comparisons

Table S46: Paired bootstrap over leakage groups on the published split, 2,000 draws, Holm within each metric; differences are LANTERN minus baseline.

Metric	Baseline	Baseline value	Diff.	95% CI	Adj. p
bAcc (raw)	16-shot probe, MedSigLIP	.748	+0.069	[+0.054, +0.084]	0.009
bAcc (raw)	Linear probe, MedSigLIP	.728	+0.089	[+0.073, +0.103]	0.009
bAcc (raw)	Zero-shot, MedSigLIP	.709	+0.108	[+0.093, +0.124]	0.009
bAcc (raw)	ERM, DINOv2 LoRA	.707	+0.109	[+0.093, +0.125]	0.009
bAcc (raw)	Group DRO, ConvNeXt-T	.707	+0.110	[+0.096, +0.125]	0.009
bAcc (raw)	Fine-tune (NLL rule), ConvNeXt-T	.677	+0.140	[+0.123, +0.156]	0.009
bAcc (raw)	CORAL, ConvNeXt-T	.676	+0.140	[+0.123, +0.157]	0.009
bAcc (raw)	SWAD, ConvNeXt-T	.672	+0.145	[+0.127, +0.161]	0.009
bAcc (raw)	Fine-tune, Swin-T	.655	+0.162	[+0.143, +0.179]	0.009
bAcc (raw)	WiSE-FT, SigLIP2-L	.634	+0.182	[+0.164, +0.200]	0.009
bAcc (raw)	MLP probe, DINOv2-B	.598	+0.218	[+0.198, +0.237]	0.009
bAcc (raw)	CoOp ctx-init, BiomedCLIP	.497	+0.320	[+0.301, +0.338]	0.009

Table S46: Paired bootstrap over leakage groups on the published split, 2,000 draws, Holm within each metric; differences are LANTERN minus baseline.

Metric	Baseline	Baseline value	Diff.	95% CI	Adj. p
bAcc (raw)	CoOp + LP, SigLIP-B	.496	+0.321	[+0.303, +0.339]	0.009
bAcc (raw)	Tip-Adapter-F, BiomedCLIP	.495	+0.322	[+0.301, +0.343]	0.009
bAcc (raw)	CoOp, SigLIP-B	.491	+0.325	[+0.307, +0.344]	0.009
bAcc (raw)	Tip-Adapter, BiomedCLIP	.478	+0.339	[+0.318, +0.359]	0.009
ECE (TS)	Zero-shot, MedSigLIP	.032	-0.012	[-0.025, +0.005]	0.221
ECE (TS)	16-shot probe, MedSigLIP	.044	-0.024	[-0.032, -0.009]	0.009
ECE (TS)	Linear probe, MedSigLIP	.079	-0.060	[-0.071, -0.038]	0.009
ECE (TS)	Group DRO, ConvNeXt-T	.091	-0.072	[-0.084, -0.052]	0.009
ECE (TS)	Tip-Adapter, BiomedCLIP	.091	-0.072	[-0.083, -0.052]	0.009
ECE (TS)	ERM, DINOv2 LoRA	.115	-0.096	[-0.108, -0.075]	0.009
ECE (TS)	Fine-tune (NLL rule), ConvNeXt-T	.133	-0.114	[-0.126, -0.092]	0.009
ECE (TS)	CORAL, ConvNeXt-T	.136	-0.116	[-0.130, -0.094]	0.009
ECE (TS)	SWAD, ConvNeXt-T	.139	-0.119	[-0.133, -0.096]	0.009
ECE (TS)	Tip-Adapter-F, BiomedCLIP	.147	-0.127	[-0.142, -0.108]	0.009
ECE (TS)	CoOp ctx-init, BiomedCLIP	.150	-0.131	[-0.145, -0.107]	0.009
ECE (TS)	WiSE-FT, SigLIP2-L	.168	-0.149	[-0.163, -0.126]	0.009
ECE (TS)	Fine-tune, Swin-T	.171	-0.151	[-0.165, -0.128]	0.009
ECE (TS)	MLP probe, DINOv2-B	.172	-0.152	[-0.166, -0.127]	0.009
ECE (TS)	CoOp, SigLIP-B	.175	-0.156	[-0.171, -0.131]	0.009
ECE (TS)	CoOp + LP, SigLIP-B	.176	-0.156	[-0.172, -0.132]	0.009
NLL (TS)	16-shot probe, MedSigLIP	.644	-0.161	[-0.186, -0.136]	0.009
NLL (TS)	Zero-shot, MedSigLIP	.690	-0.208	[-0.227, -0.188]	0.009
NLL (TS)	Linear probe, MedSigLIP	.694	-0.212	[-0.248, -0.173]	0.009
NLL (TS)	Group DRO, ConvNeXt-T	.926	-0.444	[-0.511, -0.377]	0.009
NLL (TS)	ERM, DINOv2 LoRA	.944	-0.462	[-0.526, -0.404]	0.009
NLL (TS)	CORAL, ConvNeXt-T	1.005	-0.523	[-0.589, -0.460]	0.009
NLL (TS)	Tip-Adapter, BiomedCLIP	1.020	-0.537	[-0.567, -0.506]	0.009
NLL (TS)	Fine-tune (NLL rule), ConvNeXt-T	1.036	-0.553	[-0.624, -0.485]	0.009
NLL (TS)	SWAD, ConvNeXt-T	1.056	-0.574	[-0.647, -0.502]	0.009
NLL (TS)	Tip-Adapter-F, BiomedCLIP	1.099	-0.617	[-0.652, -0.581]	0.009
NLL (TS)	Fine-tune, Swin-T	1.174	-0.691	[-0.766, -0.619]	0.009
NLL (TS)	CoOp + LP, SigLIP-B	1.180	-0.698	[-0.752, -0.644]	0.009
NLL (TS)	CoOp, SigLIP-B	1.180	-0.698	[-0.751, -0.645]	0.009
NLL (TS)	CoOp ctx-init, BiomedCLIP	1.189	-0.707	[-0.765, -0.649]	0.009
NLL (TS)	WiSE-FT, SigLIP2-L	1.243	-0.761	[-0.837, -0.684]	0.009
NLL (TS)	MLP probe, DINOv2-B	1.259	-0.777	[-0.873, -0.681]	0.009

S10 Reproducibility

Code, saved result tables and this document are archived at `https://doi.org/10.5281/zenodo.23911966` (GitHub: `https://github.com/dangaura-tech/fundusshift`). The README maps each result to its script and table file. The evaluation protocol, including the selection rules written before each later attempt, is `docs/protocol.md`. Images are obtained from their original publishers; split tables are available from the author on request.

Table S26. Test-time adaptation per held-out source: test bAcc for none, BatchNorm statistics, TENT and T3A (K = 20, 100); Chákṣu bAcc for none, TENT and T3A-100.

Setting	Model	None	BN	TENT	T3A-20	T3A-100	Ch. none	Ch. TENT	Ch. T3A
Fine-tune	swin_t	.655	–	.655	.666	.668	.298	.297	.309
Fine-tune	resnet50	.661	.598	.598	.632	.641	.417	.363	.306
Group DRO	convnext_t	.707	–	.709	.705	.703	.417	.430	.358
ERM	dinov2_lora	.708	–	.710	.644	.662	.420	.421	.298
16-shot probe	medsiglip	.748	–	–	.659	.683	.675	–	.496
Linear probe	medsiglip	.728	–	–	.679	.679	.347	–	.271

Table S27. Prior correction on top of TS: test bAcc with EM (Saerens et al.) pooled or per source, and with true priors pooled or per source (oracle, uses test labels); ECE for TS, EM per source and oracle per source.

Setting	Model	TS	EM pooled	EM per src.	True pooled	Oracle per src.	ECE TS	ECE EM	ECE oracle
Zero-shot	medsiglip	.709	.540	.635	.709	.863	.032	.287	.033
CoOp	siglip_b16	.491	.398	.457	.513	.737	.175	.433	.050
CoOp ctx-init	biomedclip	.497	.416	.438	.495	.754	.150	.389	.040
CoOp + LP	siglip_b16	.496	.394	.450	.520	.736	.176	.445	.053
16-shot probe	medsiglip	.748	.683	.787	.769	.853	.044	.185	.051
Linear probe	medsiglip	.728	.778	.835	.778	.853	.079	.120	.081
MLP probe	dinov2_b14	.598	.610	.618	.621	.821	.172	.226	.070
Tip-Adapter-F (acc. rule)	biomedclip	.495	.407	.434	.511	.739	.147	.389	.047
Tip-Adapter (acc. rule)	biomedclip	.478	.333	.495	.460	.779	.091	.292	.120
Tip-Adapter-F	biomedclip	.495	.407	.434	.511	.739	.147	.389	.047
Tip-Adapter	biomedclip	.478	.333	.495	.460	.779	.091	.292	.120
Fine-tune	swin_t	.655	.657	.661	.675	.827	.171	.226	.099
Fine-tune, NLL rule	convnext_t	.677	.687	.710	.707	.842	.133	.182	.073
CORAL	convnext_t	.676	.687	.704	.704	.851	.136	.172	.069
SWAD	convnext_t	.672	.691	.704	.706	.851	.139	.166	.060
Group DRO	convnext_t	.707	.720	.723	.733	.857	.091	.188	.051
ERM	dinov2_lora	.707	.655	.726	.727	.826	.115	.218	.104
WiSE-FT	siglip2_l16	.634	.598	.627	.669	.813	.168	.281	.103

Table S29. CLAHE against unprocessed images, same seeds and hardware.

Setting	Config	Preprocessing	Val	Test	Eq-src	Chákṣu	ECE TS	Rec. G
Fine-tune	convnext_t_ce	none	.886	.668	.612	.273	.133	.385
Fine-tune	convnext_t_ce	CLAHE	.876	.653	.604	.234	.132	.293
Fine-tune	swin_t_mbls	none	.893	.655	.600	.298	.171	.411
Fine-tune	swin_t_mbls	CLAHE	.882	.650	.608	.283	.152	.351
Linear probe	medsiglip_class_ls	none	.920	.728	.654	.347	.079	.488
Linear probe	medsiglip_class_ls	CLAHE	.911	.788	.702	.400	.041	.663
16-shot probe	medsiglip_none_ce	none	.802	.748	.683	.675	.044	.783
16-shot probe	medsiglip_none_ce	CLAHE	.790	.747	.693	.678	.033	.804

Table S30. Referral rules at a threshold for 80% coverage on Cal-B: maximum softmax (MSP), KNN distance in feature space, or both. Open: share of the 861 out-of-scope images accepted.

Setting	Model	Score	Test cov.	Test risk	Cal-B cov.	Open	G→N
Fine-tune	swin_t	MSP	.694	.250	.801	.666	.147
Fine-tune	swin_t	KNN	.415	.084	.801	.165	.007
Fine-tune	swin_t	MSP+KNN	.484	.113	.802	.252	.020
Group DRO	convnext_t	MSP	.738	.182	.801	.602	.089
Group DRO	convnext_t	KNN	.400	.076	.801	.101	.010
Group DRO	convnext_t	MSP+KNN	.487	.094	.802	.187	.026
ERM	dinov2_lora	MSP	.715	.223	.801	.610	.067
ERM	dinov2_lora	KNN	.549	.266	.801	.216	.113
ERM	dinov2_lora	MSP+KNN	.628	.221	.804	.290	.092
Linear probe	medsiglip	MSP	.610	.129	.801	.337	.061
Linear probe	medsiglip	KNN	.419	.227	.801	.059	.093
Linear probe	medsiglip	MSP+KNN	.421	.153	.802	.074	.057
16-shot probe	medsiglip	MSP	.753	.195	.801	.679	.012
16-shot probe	medsiglip	KNN	.419	.223	.801	.059	.027
16-shot probe	medsiglip	MSP+KNN	.483	.183	.802	.105	.020

Table S31. Out-of-scope detection AUROC from penultimate features of trained models (SWAD rows omitted: its saved checkpoint holds pre-averaging weights).

Setting	Model	MSP	MaxLogit	Energy	KNN	Mahalanobis
Fine-tune	swin_t	.549	.567	.572	.664	.754
Group DRO	convnext_t	.618	.610	.606	.745	.791
ERM	dinov2_lora	.613	.610	.610	.753	.754

Table S32. Multiple-choice answering by multimodal LLMs (mean over six option orders).

Model	Test bAcc	Rec. N	Rec. D	Rec. G	Chákṣu bAcc	Chákṣu Rec. G
Fundus-R1-7B	.576	.789	.795	.143	.463	.128
Qwen2.5-VL-7B-Instruct	.424	.926	.347	.000	.488	.005
llava-med-v1.5-mistral-7b-hf	.347	.905	.056	.080	.507	.037
llava-onevision-qwen2-7b-ov-hf	.407	.367	.855	.000	.096	.000

Table S33. Recall of D by DR grade on test (grade −1: Davis grading without a numeric ICDR grade).

Setting	Model	−1	1	2	3	4
Zero-shot	medsiglip	.785	.000	.734	.954	.838
CoOp	siglip_b16	.711	.444	.688	.823	.915
CoOp ctx-init	biomedclip	.808	.111	.618	.884	.925
CoOp + LP	siglip_b16	.718	.444	.700	.835	.929
16-shot probe	medsiglip	.767	.056	.841	.989	.967
Linear probe	medsiglip	.727	.000	.844	.981	.977
MLP probe	dinov2_b14	.665	.222	.737	.935	.932
Tip-Adapter-F (acc. rule)	biomedclip	.660	.167	.569	.831	.885
Tip-Adapter (acc. rule)	biomedclip	.368	.000	.122	.432	.499
Tip-Adapter-F	biomedclip	.660	.167	.569	.831	.885
Tip-Adapter	biomedclip	.368	.000	.122	.432	.499
Fine-tune	swin_t	.724	.389	.844	.974	.986
Fine-tune, NLL rule	convnext_t	.717	.389	.801	.965	.969
CORAL	convnext_t	.704	.278	.801	.966	.972
SWAD	convnext_t	.724	.278	.817	.966	.972
Group DRO	convnext_t	.712	.111	.743	.969	.952
ERM	dinov2_lora	.701	.333	.865	.980	.996
WiSE-FT	siglip2_l16	.695	.222	.755	.964	.968

Table S34. Glaucoma recall on clean test images and at the strongest blur and JPEG levels.

Setting	Model	Clean	Blur	JPEG
Fine-tune	swin_t	.419	.071	.100
Group DRO	convnext_t	.447	.026	.073
ERM	dinov2_lora	.598	.438	.358
16-shot probe	medsiglip	.784	.267	.870
Linear probe	medsiglip	.491	.144	.086

Table S35. Complete specification of LANTERN.

Encoder	MedSigLIP (patch 14, 448 px, 27 layers, width 1152), frozen
Head input	Last hidden state, 32×32 tokens pooled to 14×14; fixed random orthonormal map to 384 dimensions (seed 0); grid mean
Standardization	Per dimension, training-split mean and standard deviation
Projection	Within-class between-site means, at most 32 directions, fitted per head on its training rows
Heads	One linear head per development source, trained without it; 1,155 parameters each
Training	Label smoothing 0.1; class-balanced sampling; AdamW, lr 3×10^{-4}, wd 10^{-2}; batch 64; 12 epochs; best epoch by validation bAcc on the head's own sites
Rescaling	Centre per image; divide by the standard deviation over validation rows
Text	Mean of the three banks of Table S5, on the pooled image embedding
Fusion	0.5 heads + 0.5 text; TS on Cal-A
Seeds	0, 1, 2 on the published split; one per rotation

Table S38. LANTERN on each held-out development site.

Held-out site	bAcc	Recall G	False G
Bangladesh Eye	.751	.716	.149
DeepDRiD	.776	–	.081
GRAPE	.812	.812	–
IDRiD	.758	–	.026
Jichi	.767	–	.101
OIA-DDR	.730	–	.202
SUSTech-SYSU	.946	–	.040

Table S39. Designs tried and not retained (development). Reference: pooled-feature pipeline 0.791; single cross-entropy head 0.675.

Experiment	bAcc	Reading
Batch network predicting temperature and class bias (1 seed)	.539	Below the raw head (.666)
EM prior shift with TS (1 seed)	.523	Unreliable with few classes per source
Polar Fourier features around the disc	.702	Below the pair loss alone (.723)
Site prior, κ from 30 to 3000	.693–.717	No gain
Full residual corrector over logits	.667	Recall G fell to 0
Corrector with fusion weights only	.781	ECE .057, bAcc did not rise
RPAttn head on three token depths (1 seed)	.664	Recall G .008; with text .690
RPAttn + MixStyle (1 seed)	.614	Recall G .012; with text .688
N/D from RPAttn, G from low-capacity branch (1 seed)	.788	.793 for the pooled pipeline on that seed
Late fusion of nine encoder inputs	.765	Other encoders lowered the score
Greedy nested encoder selection	.789	Mostly kept MedSigLIP alone
Magnified optic-disc view with whole image	.768	Recall G up (.767), false G up (.135)
Image-conditioned fusion gate	.763	Below a plain average
Global class offset chosen on other sites	.679	GRAPE fell to .006
ConvNeXt-T end-to-end with pair loss (1 seed)	.622	GRAPE .004

Table S40. Held-out development sites for the pooled-feature pipeline: head, text and fusion bAcc, recalls, and images wrong under all nine encoder inputs.

Held-out site	*n*	Head	Text	Fusion	N	D	G	All wrong
Bangladesh Eye	3,153	.651	.732	.753	.761	.963	.536	.039
OIA-DDR	11,269	.660	.738	.735	.685	.786	–	.137
DeepDRiD	1,804	.803	.721	.795	.682	.908	–	.111
GRAPE	528	.644	.750	.741	–	–	.741	.104
IDRiD	462	.757	.752	.751	.527	.974	–	.100
Jichi	8,475	.779	.736	.803	.824	.781	–	.103
SUSTech-SYSU	1,097	.975	.889	.957	.938	.975	–	.009

Table S41. Normal images predicted G, by disc side (automatic, label-free disc location), held-out site.

Held-out site	Disc left	Disc right
OIA-DDR	.386	.191
Bangladesh Eye	.141	.023
IDRiD	.119	.045
SUSTech-SYSU	.086	.019
Jichi	.067	.054
DeepDRiD	.096	.086

Table S43. Post-processing checked on held-out development sites, fitted on calibration rows of the other sites.

State	ECE	NLL	Cov 80 cov.	Cov 80 risk	Risk 10 risk	LAC cov.	LAC size	Class-LAC	APS cov./size
Raw	.148	.576	.720	.105	.094	.866	1.18	.846	.930/1.62
TS	.052	.502	.719	.104	.096	.867	1.18	.842	.946/1.78
FTS	.053	.505	.720	.106	.096	.867	1.18	.843	.919/1.61

Table S44. Label scope: development (scope-aware bAcc counts a DR-only normal predicted G as correct) and the 50 rotations.

Arm	bAcc	Scope-aware bAcc	ECE	Recall G	N→G, DR-only	N→G, full scope
exact	.792	.844	.151	.764	.145	.322
scope	.776	.864	.136	.811	.244	.255
exact (rotations)	.771	.801	.033	.667	–	.248
scope (rotations)	.761	.831	.040	.797	–	.443

Table S47. Configurations with development bAcc of at least 0.791, all evaluated on the published split.

Configuration	Dev	Test bAcc	Eq-src	Recall G	NLL (TS)	Chákṣu
Mirror, weight decay 10^{-3}	.800	.809	.726	.696	.503	.714
Mirror, weight decay 10^{-2}	.799	.808	.727	.708	.502	.726
Mirror, weight decay 10^{-4}	.795	.807	.724	.681	.511	.703
Cross-site pair loss	.794	.809	.717	.722	.507	.694
Same-site pair loss	.792	.791	.696	.670	.524	.713
Tokens, weight decay 10^{-2} (LANTERN)	.792	.817	.724	.765	.482	.731
Pooled feature, weight decay 10^{-4}	.791	.799	.706	.687	.520	.684
Pooled feature + tokens	.791	.812	.719	.719	.493	.701
Disc side canonicalised	.791	.805	.733	.679	.511	.637
Mirrored text only	.791	.800	.714	.679	.512	.704